in 2026

Agentic AI Interview Preparation

Crack Big Tech Interviews

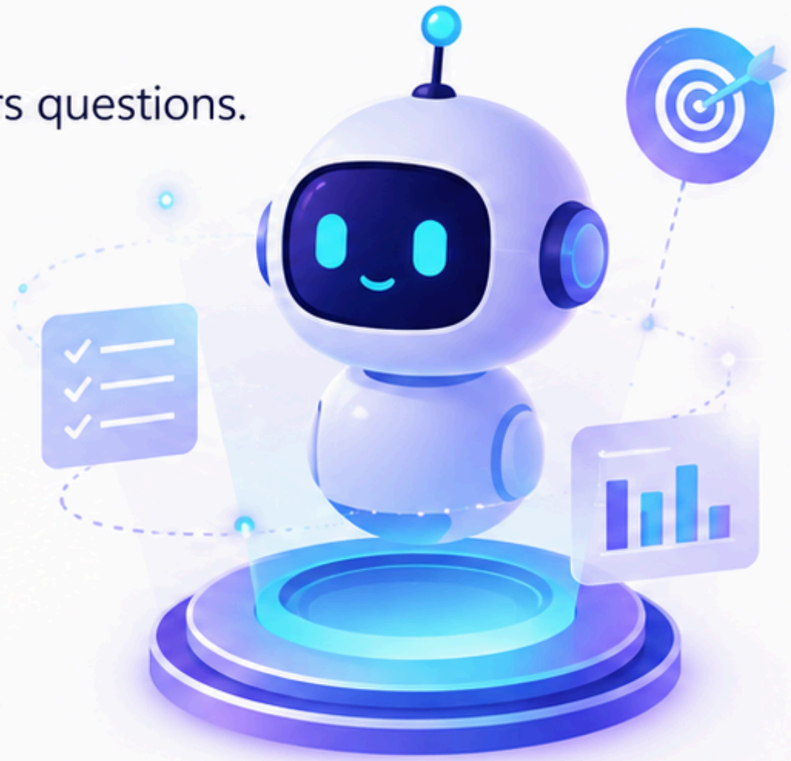
Q1. What is Agentic AI and how is it different from Generative AI?

Agentic AI refers to AI systems that can **autonomously plan, decide, and execute actions** to achieve a defined goal **without continuous human intervention.**

Unlike **traditional AI** that only responds to prompts and answers questions.

Agentic AI can:

- Plan
- Autonomous decision-making
- Ability to use tools and external systems
- Learn from past experiences to complete tasks independently.



Q1. What is Agentic AI and how is it different from Generative AI?

Generative AI

Example:

You ask, *"Draft a meeting agenda"*

Meeting Agenda

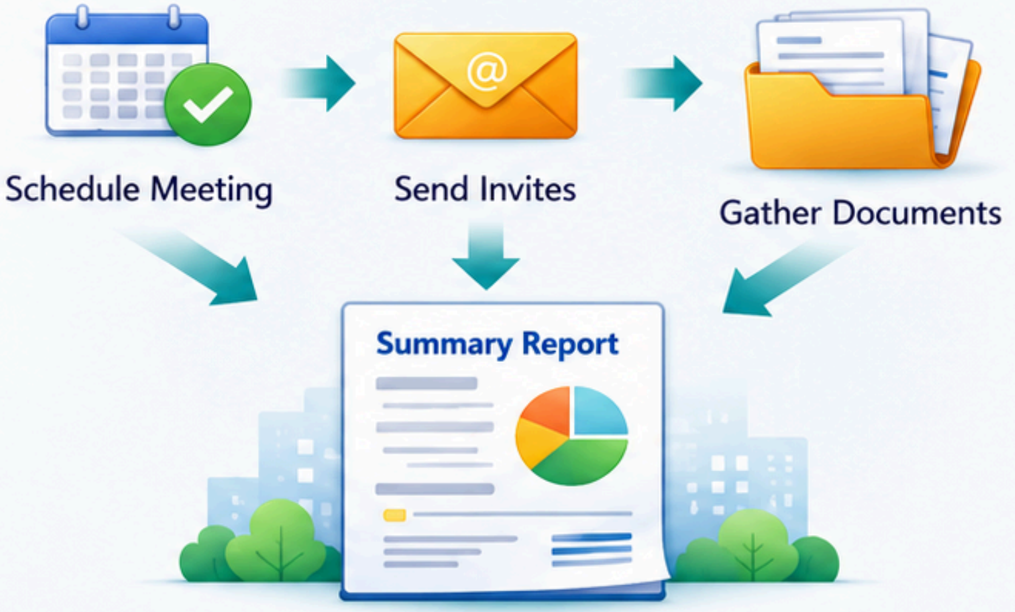
1. Welcome & Introductions
2. Project Updates
3. Discussion Topics
4. Next Steps

Generative AI Stops

Agentic AI

Example:

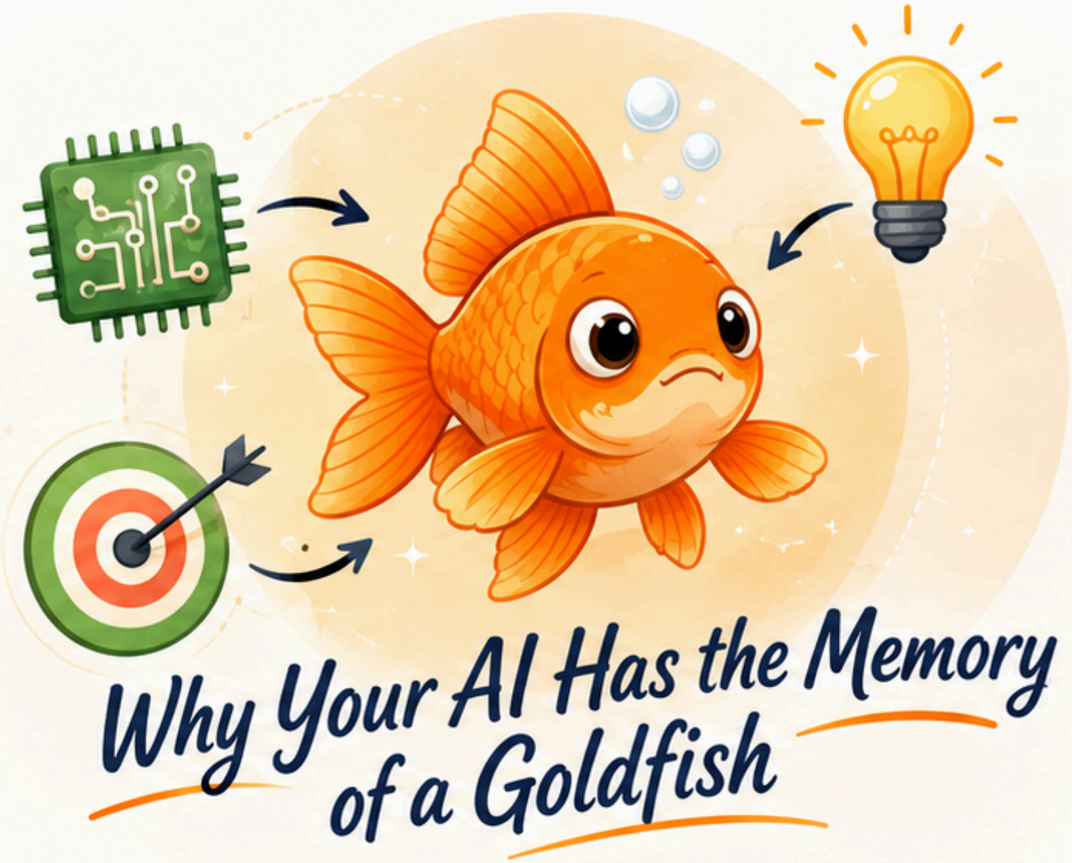
Agentic AI schedules the meeting, sends invites, gathers documents, and drafts a summary report.



Q2. What are the core components of an AI Agent?

1. **Reasoning (Core LLM engine):** Decides the next action based on goals and logic.
2. **Planning:** Breaks a high-level goal into ordered sub-tasks.
3. **Memory:** Stores past actions, user preferences, and contextual information.
4. **Perception (Input layer):** Gathers data to understand the environment.
5. **Action (Execution layer):** Uses tools like APIs and databases to perform real-world tasks.

Q3. Why Memory is Critical for AI Agents, and what are the different types of Agent Memory?



Memory is what provides an agent with **context, continuity, and learning ability,** separating it from a simple stateless chatbot.

Without memory, every interaction starts from scratch, like a goldfish forgetting what happened a few seconds ago. This is called **“Goldfish Syndrome.”**

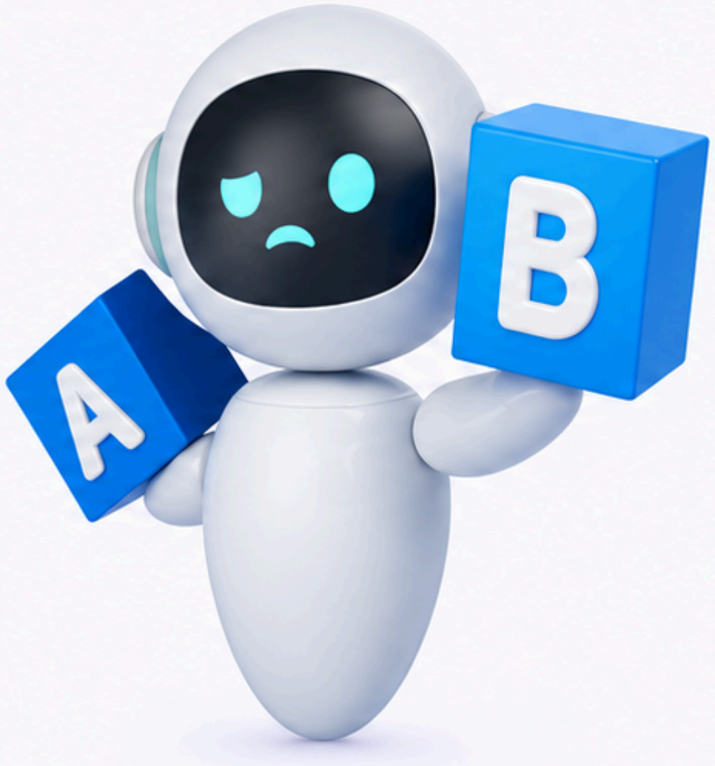
Q3. Why Memory is Critical for AI Agents, and what are the different types of Agent Memory?

1. **Short-term Memory:** Tracks context within a single task or conversation (context window).
2. **Long-term Memory:** Stores persistent knowledge and experiences across sessions, usually via vector databases.
3. **Working Memory:** Acts as the scratchpad for active reasoning and multi-step decision-making during a task.
4. **Episodic Memory:** Remembers specific past events like system failures, helping improve future decisions.

Q4. What is the difference between Reactive and Deliberative AI Agents?

Reactive Agents: Reactive Agents are agents that respond instantly to specific current situations happening around them.

They do not think in terms of future or long-term perspective. They are fast and efficient, making them useful in simple, predictable environments where quick responses matter.



Example:

A thermostat that turns on the AC the moment it hits 25°C.

Q4. What is the difference between Reactive and Deliberative AI Agents?

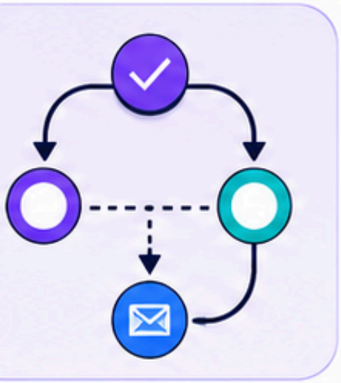
Deliberative Agents: Deliberative Agents use a reasoning engine to evaluate their environment, maintain an internal state, and plan multiple steps ahead.

They are “deliberate” because they consider the consequences of their actions before executing them. Best suited for complex long-term planning and multi-step decision-making.

Example:

A travel agent AI that evaluates multiple flight and hotel combinations against a budget before proposing a plan.

Q5. How do you choose the right framework for Agentic AI: LangGraph, CrewAI, n8n, or AutoGen?

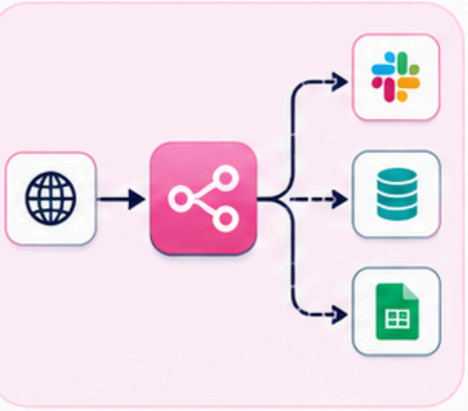


LangGraph:

Best for step-by-step sequences where the AI follows a specific path to use tools like Search → Summarize → Email.

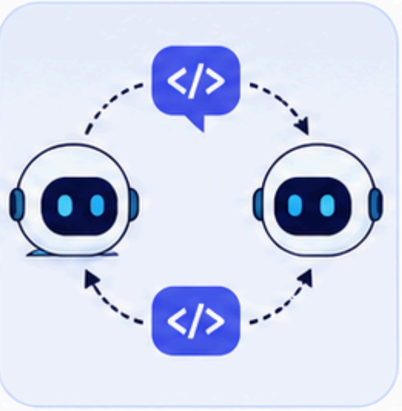
CrewAI:

Best for role-based collaboration, where multiple specialized agents work together like a “crew”.



n8n:

Best for workflow automation and integrations, where the agent connects apps, APIs, and business processes.



AutoGen:

Best for open-ended problem solving where agents interact and communicate to solve complex tasks like debugging or coding.

Q6. What is the approach to building an AI Agent?

1. **Define the Goal:** Clearly identify the specific outcome like automation, or research.
2. **Design the Architecture:** Choose the core LLM (Brain) and reasoning layer like ReAct.
3. **Implement Memory:** Add short-term memory and long-term memory using vectorDB.
4. **Integrate Tools:** Connect APIs and utilities that allow real-world actions.
5. **Set Up Feedback Loops:** Add evaluation systems and Human-in-the-Loop (HITL).
6. **Test and Deploy:** Run in sandbox, validate safety, then deploy to production.

Q7. What is the Retrieval-Augmented Generation (RAG) pipeline and how does it improve AI agents?

RAG

Retrieval-Augmented Generation (RAG) is a technical architecture that provides an AI agent with a dynamic knowledge base beyond its original training data.

It enables the agent to:

- Access external or private data
- Reduce dependence on static training knowledge
- Generate responses grounded in real information

Q8. What is the Human-in-the-Loop (HITL) approach?

Human-in-the-Loop (HITL) is a governance framework where **human intelligence and guidance** are integrated into the AI's decision cycle.

The AI performs most of the work, but a human is actively involved to:

- Review outputs
- Guide decisions
- Correct or validate actions

This makes the system more **safe, reliable, and trustworthy** than fully autonomous AI.

Q9. What is emergent behavior and how do you manage it?

Emergent behavior is when a group of AI agents develops complex, collective patterns that were never specifically programmed.

It follows the principle that **“the whole is greater than the sum of its parts”**, where simple individual actions escalate into sophisticated group intelligence.

Q10. How would you design a multi-layer guardrail system to ensure agent safety and performance?

The 5-Layer Guardrail Architecture

1. **Input Rails:** Filters prompt injection, and personal data before reaching the LLM.
2. **Dialog Rails:** Ensures conversation stays within predefined safe paths and redirects restricted attempts.
3. **Retrieval Rails:** Validates RAG data for relevance, safety, and access control compliance.
4. **Execution Rails (Kill-Switch):** Verifies tool actions before execution using intent verification.
5. **Output Rails:** Final check for hallucinations, data leaks, or unsafe responses before output is returned.

Q11. How do you implement rollback and error handling in autonomous agents?

5-Step Error Handling Workflow

1. **Validation:** Run assertions after every action to verify the outcome.
2. **Detection:** Identify failures through errors, or unexpected outputs.
3. **Action Formulation:** Apply fallback strategies like retrying or modifying the task logic.
4. **Logging:** Maintain a complete decision (trace logs) for debugging and improvement.
5. **Human-in-the-Loop:** Escalate high-risk failures for human review and intervention.

Q12. Suppose your agent system costs $8,000/month. How would you reduce it by half without degrading quality?

Cost Optimization Techniques

1. **Model Tiering:** Use a router to assign simple tasks to lightweight models and complex tasks to high-performance models.
2. **Semantic Caching:** Store previous answers in a vector cache and reuse them.
3. **Optimize Workflows:** Break large tasks into smaller steps to avoid redundancy and token waste.
4. **Batch Processing:** Use batch APIs for non-urgent tasks to benefit from discounted pricing.
5. **Monitor & Limit:** Track real-time costs and prevent infinite loops using hard-stop limits.

Q13. How do you manage context windows to optimize token usage and performance?

Context Optimization Techniques

1. **Context Distillation:** Uses a small model to remove noise, metadata, and irrelevant text.
2. **Recursive Summarization:** Compresses older conversation into short summaries while preserving intent.
3. **Dynamic RAG (Retrieval):** Fetches only the most relevant chunks instead of full documents.
4. **Sliding Window Strategy:** Drops oldest messages while preserving system prompt and recent context.
5. **Task Segmentation:** Breaks large goals into micro-tasks to avoid context overload.

Q14. How do you transform an LLM into an autonomous, goal-directed AI agent?

5-Step Transformation Workflow

1. **Planning:** Uses ReAct (Reason + Act) to create and update a dynamic to-do list.
2. **Memory:** Short-term scratchpad for current tasks and long-term vector memory for past experiences.
3. **Tools:** Connects APIs, databases, and external systems to perform real-world actions.
4. **Observation:** Feeding tool outputs back into the LLM so it can evaluate results.
5. **Guardrails:** Reflexion layer ensures alignment with the original goal and prevents infinite loops.

Q15. What security risks should be considered when deploying autonomous agents?

The 5 Critical Security Risks

1. **Indirect Prompt Injection:** Hidden malicious instructions inside external data sources like websites or emails.
2. **Privilege Escalation:** Giving excessive permissions (e.g., admin access) leading to potential system damage.
3. **Data Exfiltration:** Accidental leakage of sensitive data into logs or external tools.
4. **Denial of Wallet:** Infinite loops causing uncontrolled API calls and high financial cost.
5. **Lack of Human Approval:** Executing irreversible actions without human confirmation.